

Rolling out NDN for DDoS Mitigation

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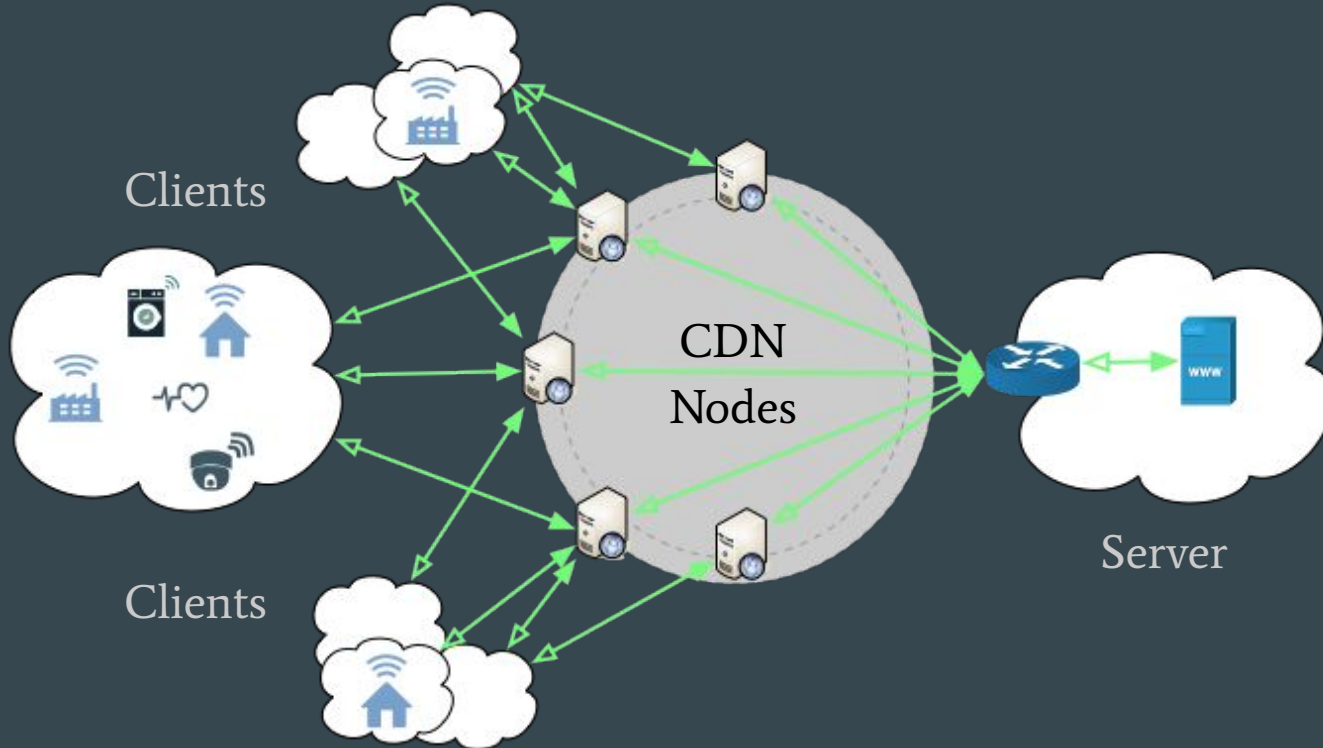
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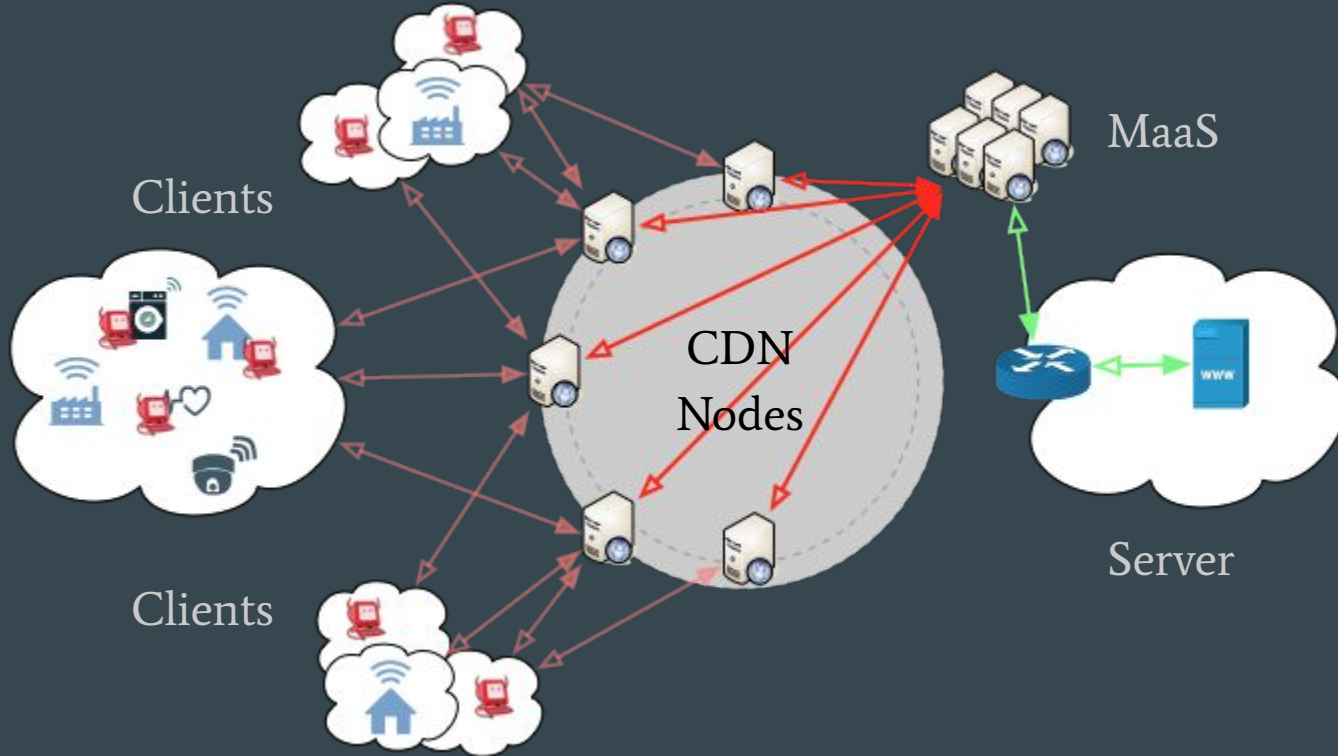
DDoS can be a “Killer App”

- DDoS has plagued the Internet for decades
- It can be a killer app to push architectural changes
- Our candidate NDN has intrinsic DDoS mitigation properties
 - Off by default
 - Stateful forwarding
 - Name provides traffic insights
 - No reflection attack
 - ...

How Today's CDN Works



When DDoS happens, MaaS is involved



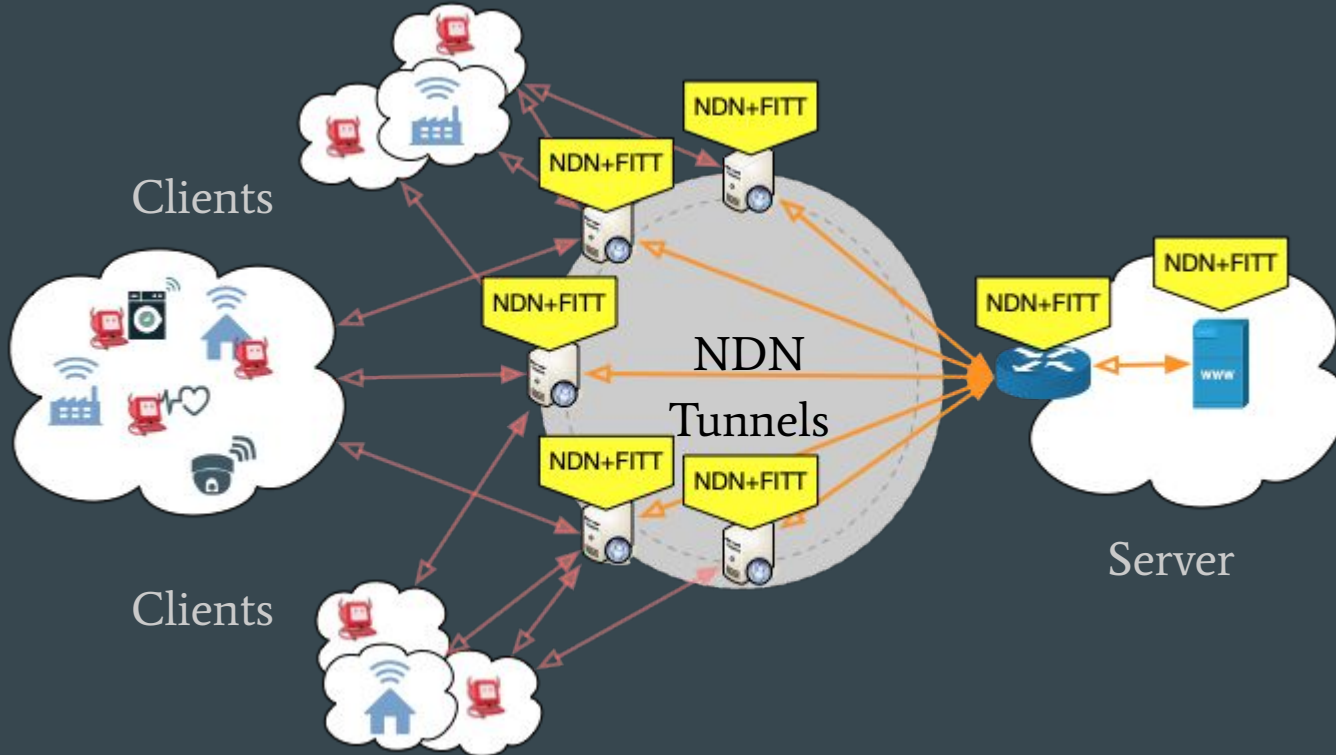
Is Today's DDoS Mitigation Good Enough?

- CDN providers have already start providing DDoS mitigation services with their existing CDN infrastructure
- However...
 - Deep packet inspection (DPI) with TLS traffic is expensive (thus usually require centralized MaaS)
 - Extra MaaS infra is usually needed
 - Blackholing or backhauling traffic cause collateral damages to good clients
 - Slow reaction

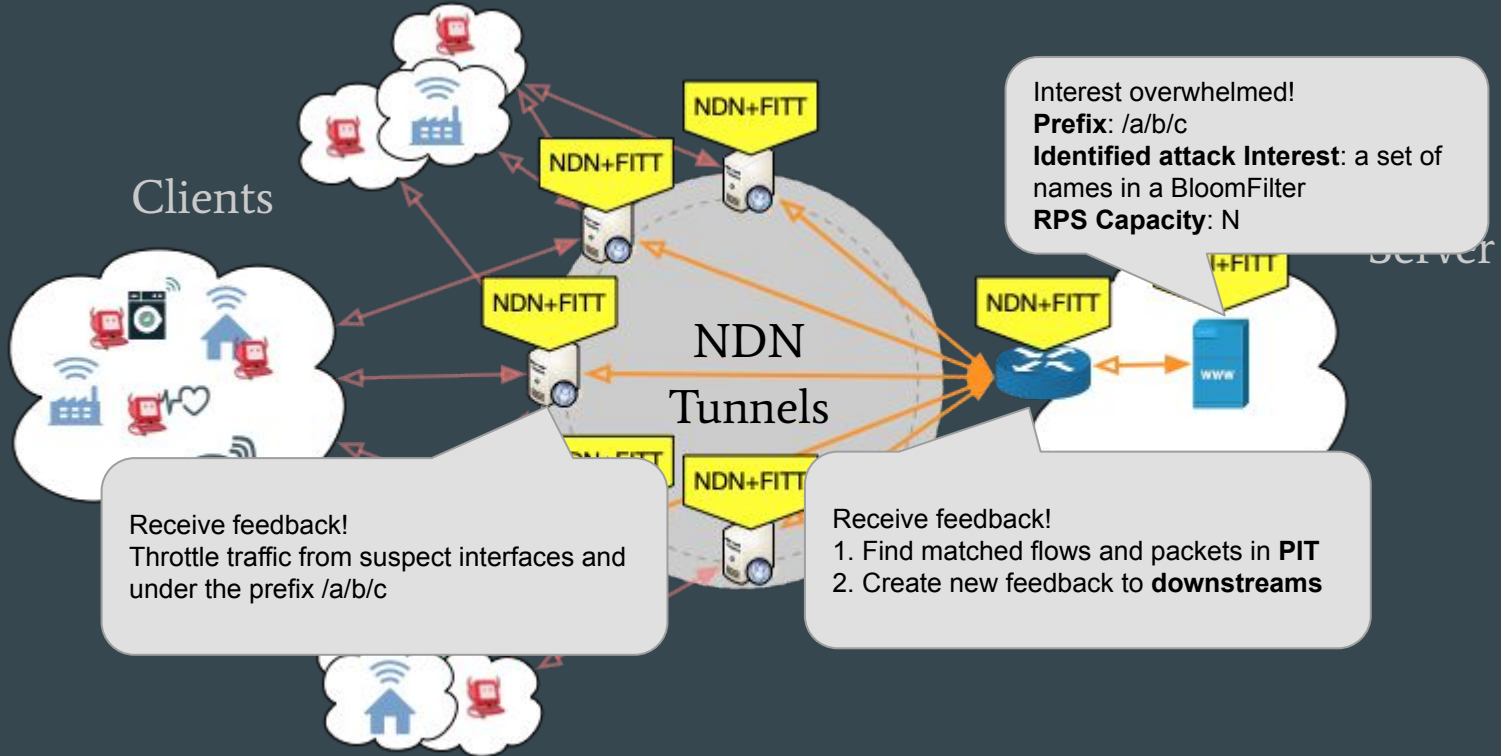
NDN + FITT for DDoS Mitigation

- Fine-grained Interest Traffic Throttling (FITT) is our NDN-based DDoS mitigation system that utilizes NDN stateful forwarding
 - Refer to our TR: <https://arxiv.org/abs/1902.09033>
- This talk, focus on how to do incremental deployment of NDN and FITT using today's CDN boxes and analyze the **incentives**
 - The same approach works for other NDN/ICN DDoS mitigation mechanisms as well :p
- We show NDN + FITT can help today's CDN to serve better -- Better CDN and DDoS mitigation services with smaller budget

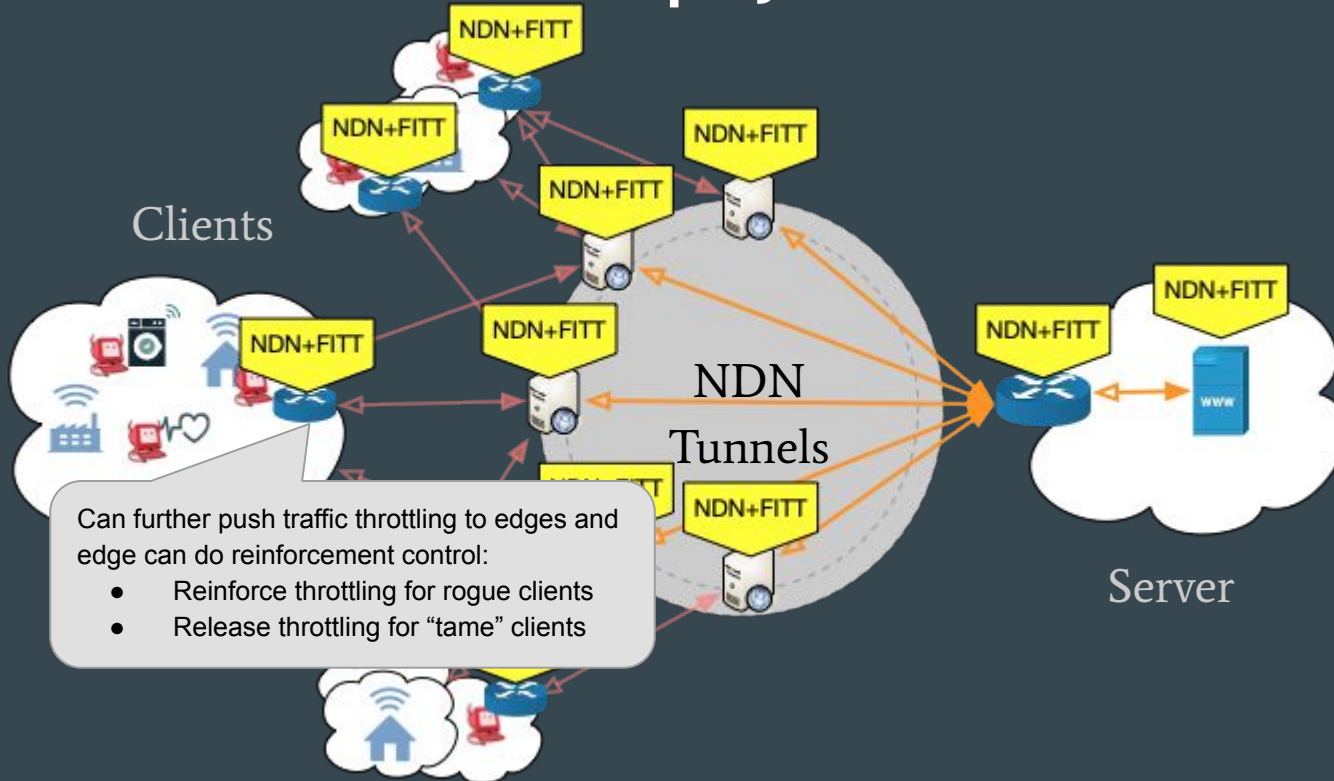
Incremental Deployment of NDN+FITT



Mitigate DDoS with NDN + FITT

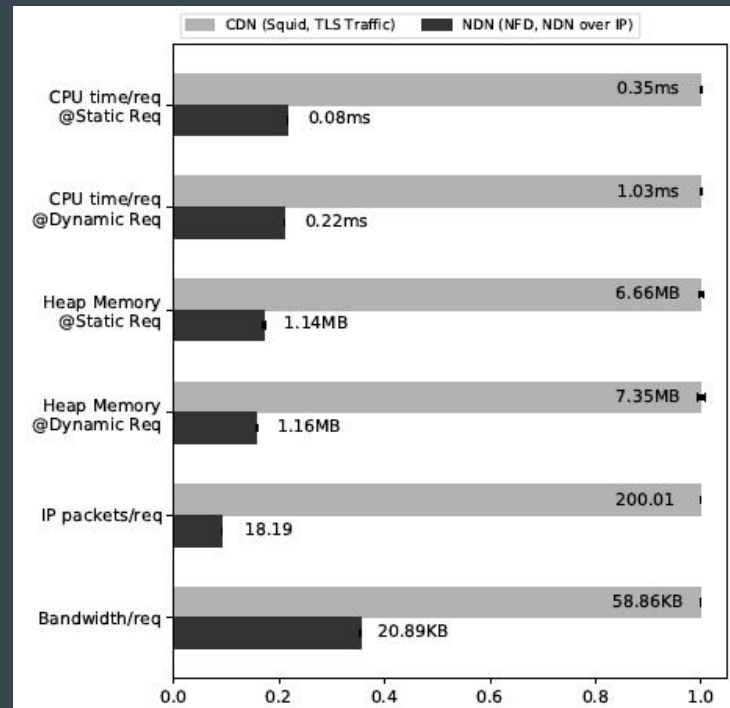


Incremental Deployment of NDN+FITT



Incentives of Deployment

- NDN + FITT does **both CDN** and **DDoS mitigation**
- No need for additional hardware (MaaS)
- A smaller resource footprint compared with current practice (TLS termination + DPI + scrubbing)
 - Proved by evaluating real deployment: NDN + FITT v.s. Squid for CDN and TLS termination (didn't even add scrubbing and TLS forwarding)
 - NDN + FITT: $\frac{1}{5}$ CPU & MEM, $\frac{1}{3}$ Bandwidth



Desired Features of FITT over NDN

- Fine-grained throttling per prefix at real attack origin => minimal collateral damage to benign traffic
- Distributed throttling
- Fast reaction: a half RTT from server to edge to start throttling
- Generality: mitigate all types of NDN Interest floodings
 - Compared with existing Interest flooding mitigation approaches focusing on prefix+random Interest attack

FITT is enabled by NDN's architectural properties

- Names provide more insights of traffic than IP + port #, thus fine-grained traffic classification
- Stateful forwarding allow accurate traffic traceback

Takeaways

- The architecture plays an important role in DDoS resiliency and mitigation
 - TCP/IP v.s. NDN
- NDN/FITT can be incrementally deployed over today's CDN boxes to facilitate DDoS mitigation
 - Future work: further investigations and experiments

Thank you!
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